

Presentation Script

SHAR — Self-Supervised Learning for Activity Recognition on Imbalanced Datasets

with the Health Tracking Dashboard

A section-by-section guide to what to say when demonstrating the project. Each part has a short explanation, an exact line you can say, and the key numbers to point at.

If you have only 30 seconds — the one-line pitch

“We train an activity-recognition model on smartphone sensor data. Two problems: the classes are imbalanced, and labels are expensive. We solve the first with **iSMOTE** balancing and the second with **self-supervised pre-training** — so using only **25% of the labels** we reach **87.3% accuracy**, and even the rare classes score as well as the common ones. Then we turn those predictions into a real **health dashboard** — calories, sleep, falls, heart rate.”

Explain the problem before opening the app.

Say: "Human Activity Recognition means using a phone or smartwatch's motion sensors to detect what a person is doing — walking, sitting, climbing stairs. It powers fitness bands and fall-alert devices. Two real problems make it hard: first, the activity data is **imbalanced** — people sit far more than they climb stairs, so the model rarely sees the rare activities. Second, **labelling** sensor data by hand is slow and expensive. This project solves both, and then shows the payoff as a practical health-monitoring dashboard."

The three ideas you are demonstrating:

- **iSMOTE** — an improved way to synthetically balance the rare classes without adding noise.
- **Self-supervised pre-training** — the model learns the structure of the signals with *no labels*, so only a few labels are needed afterwards.
- **The application** — the recognised activity drives calorie, step, sleep, fall and heart-rate insights.

Then say: "The dashboard is built as two steps. Step 1 is where we choose a dataset and balance it. Step 2 unlocks all the health features on top of that balanced data. Let me walk through it."

1

Step 1 — Dataset & Balancing

The landing page. This is where the 'imbalanced datasets' part of the title is proven.

1a. Choosing a dataset

Say: "The project ships with two real research datasets. **UCI HAR** — 6 activities, 30 people, a waist phone with 9 sensor channels. And **WISDM** — 18 activities, 51 people, a phone accelerometer. I'll use UCI HAR for the demo. When I select it, the app loads the real training data straight from disk."

1b. The original (imbalanced) distribution

Say: "Here is the raw class distribution. Notice it is uneven — the imbalance ratio tells us the biggest class has this many times more samples than the smallest. A model trained on this would get good at the common activities and poor at the rare ones."

1c. Running iSMOTE

Say: "Now I click **Run iSMOTE**. Ordinary SMOTE creates synthetic samples by interpolating between minority examples, but it often creates noisy points that overlap other classes. **iSMOTE improves this:** after generating a candidate, it checks the candidate's nearest neighbours across the *whole* dataset — if they aren't all the same class, the candidate is rejected. So we only keep clean synthetic samples."

After it finishes, point at the before/after charts:

Say: "On the left, the imbalanced original. On the right, every class is now brought up to the same height — the green portion is the synthetic samples iSMOTE added. The imbalance ratio is now 1.0. This balanced dataset is what everything else is built on."

Tip: mention the PDF export button — "the app can export this balancing summary as a report." And the 'peek at a real signal' panel shows an actual raw sensor window.

2

Dashboard — the daily overview

The first Step-2 page. A one-glance summary of a simulated day.

Say: "Once the dataset is balanced, the dashboard unlocks. This is the daily overview a user would see. The banner at the top confirms which dataset we're working on. The cards show total and active calories, steps, average heart rate, sleep hours and sleep score — all derived from the *same* simulated 24-hour day, so the numbers are consistent with each other."

Point at the two charts and the alerts:

- **24-hour activity timeline** — a colour band showing what the person did across the day (locomotion, sedentary, sport, daily living).
- **Heart-rate trend & hourly calorie burn** — both computed from that day's activity sequence.
- **Health alerts** — e.g. a sedentary warning that correctly excludes sleep, and a step-goal card.

Honest framing: "The heart rate and the daily timeline here are realistic simulations, clearly labelled as such. The genuinely model-driven part is the Activity Classifier page, which I'll finish on."

3

Calories & Steps

Shows the energy-expenditure science behind the numbers.

Say: "Calories are computed using **MET values** — Metabolic Equivalent of Task — the standard measure of how intense an activity is. One MET is resting; jogging is about seven. I can pick any activity and duration and it computes the calories for our body profile using the standard formula: $\text{MET} \times \text{weight} \times \text{hours}$."

Say: "The step counter is real signal processing — it finds peaks in the accelerometer magnitude, because each step is a bounce. The chart marks every detected step, and from the step count and cadence we estimate distance."

4

Sleep Analysis

Sleep staging from how much the body moves.

Say: "During sleep, the amount of micro-movement tells us the sleep stage — very still is deep sleep, more movement is light or REM. We measure the standard deviation of the accelerometer in each 5-minute window and classify it. The **hypnogram** shows the night moving through Deep, Light and REM stages, and we score the night out of 100 based on how close the stage mix is to a healthy ideal."

Key point to make: the stages shown are *produced by the classification algorithm* on the movement signal — not hand-drawn. The chart and the score come from the same data.

5

Fall Detection

A safety feature — threshold-based, three phases.

Say: "A fall has a recognisable signature in three phases. First **free-fall** — acceleration drops toward zero as the body drops. Then **impact** — a sharp spike when it hits the ground. Then **immobility** — the person lies still afterwards. Our detector checks all three, which is what stops normal movements from triggering false alarms."

Say: "The two tabs prove it works both ways: on a simulated fall it fires the alert and marks the impact; on a normal walking signal it correctly stays silent. In a real product this would send an alert to a carer."

6

Heart Rate & Stress

Cardiovascular metrics, plus the reverse-lookup tool you added.

Say: "This page simulates heart rate and **HRV** — heart-rate variability, which drops when the body is stressed or working hard — to give a stress estimate across activity zones."

6a. The 'What Am I Doing?' tool

Say: "This is a reverse lookup. I type in a heart rate — say 95 bpm — and it tells me the heart-rate zone and ranks which activities are most likely at that heart rate. A resting rate suggests sitting or lying; a high rate suggests stairs or jogging. It's a simple, interpretable way to connect a single number back to behaviour."

If asked about limits: *it distinguishes by intensity, so it separates rest from vigorous confidently, but two activities in the same zone (e.g. walking vs kicking) look similar by heart rate alone — the ranked bar chart shows that honestly.*

7

Activity Classifier — the research payoff

Finish here. This is the actual trained model and the evidence for the whole thesis.

7a. Trained-model performance (lead with the numbers)

Say: "This model was fine-tuned on only **25% of the labels**, yet reaches **87.3% test accuracy** and **0.873 macro-F1** on data from people it never saw in training. And these aren't typed in — they're read from the file the training pipeline writes, so they're fully reproducible."

Class	Precision	Recall	F1
Walking	0.94	0.87	0.91
Walking Upstairs	0.89	0.87	0.88
Walking Downstairs	0.86	0.96	0.91
Sitting	0.74	0.81	0.77
Standing	0.86	0.79	0.82
Laying	0.96	0.95	0.95

Say: "Point at Walking Upstairs and Downstairs — the two rarest classes in the training set. They score F1 of 0.88 and 0.91, just as good as the common classes. **That is the iSMOTE balancing doing its job** — without it, those rare classes are exactly where a model normally fails. This chart is the proof of our main claim."

7b. The t-SNE figure — what self-supervision learned

Say: "This figure is the proof for the self-supervised part. On the left, the model before training — the activities are jumbled together. On the right, after contrastive pre-training that used **zero labels** — the activities have separated into their own clusters on their own. The network discovered the structure of human activity by itself; the labels only came later to name the clusters."

7c. Live per-sample inference

Say: "I can drag this slider to any of the ~3,000 test samples. On the left is the raw sensor signal; on the right the model's confidence for each activity, with its prediction next to the true label — nothing cherry-picked."

Power move — find a wrong one on purpose:

Say: "Here it confuses Sitting and Standing — and if you look, it isn't confidently wrong, it's genuinely unsure between the two. That's expected: both are still postures that look almost identical to a waist sensor. Our errors land exactly where the physics says they should, which is a sign of an honest model."

Q

Likely questions — and your answers

Q: Why use only 25% of the labels?

A: Because labelling sensor data by hand is expensive. The whole point of self-supervised learning is to need fewer labels — we show strong accuracy with a quarter of the annotation cost.

Q: Is the test data really unseen?

A: Yes. UCI HAR is split by *person* — the test subjects never appear in training, so this measures how well it generalises to new people, not memorisation.

Q: What is iSMOTE actually doing differently from SMOTE?

A: Plain SMOTE can create synthetic points that overlap other classes and add noise. iSMOTE validates each new point against its neighbours in the whole dataset and rejects it if they aren't the same class — so the synthetic data stays clean.

Q: What is the hardest thing for the model?

A: Telling Sitting from Standing — both are still postures with almost identical waist-sensor signals (F1 0.77 and 0.82). A wrist or pocket sensor, or using the gyroscope's posture angle, would fix it.

Q: What is the Lambda layer / contrastive loss?

A: The Lambda layer is an efficient alternative to standard attention for long sensor sequences. The contrastive loss (NT-Xent) trains the model to recognise that two differently-masked copies of the same signal belong together — that's how it learns without labels.

Q: Why is WISDM's accuracy lower than UCI HAR's?

A: WISDM has 18 classes vs 6 and I only ran a short demo training on it. The pipeline is identical; a full training run brings it up. UCI HAR at 87% is the properly-trained showcase.

Q: Can you prove SSL + iSMOTE actually helps?

A: Yes — there's an ablation script that compares supervised-from-scratch vs SSL vs SSL+iSMOTE across label amounts, with error bars over several runs. It produces the comparison table directly.

Closing line: “So end-to-end — we take imbalanced, barely-labelled sensor data, balance it with iSMOTE, learn from it with self-supervision, hit 87% accuracy on unseen people, and turn every prediction into a real health insight. That's the whole pipeline in one dashboard.”